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FRANCE

Dr. Pierre Martinez
Bioinformatics of Cancer Evolutionary Dynamics

Nationality: French
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EDUCATION

- 2023 **HDR** (*Habilitation à Diriger des Recherches*), Université Claude Bernard Lyon 1.
- 2008-2011 **PhD** under the supervision of Prof. Franca Fraternali (Randall Division of Cell and Molecular Biophysics, King's College London, England).
"Computational Analysis of Transcription Factors Cooperation and Protein-Protein Interactions Structural Features". *CASE Award (EPSRC)* in collaboration with the Nestlé Research Centre (Dr James Holzwarth, Lausanne, Switzerland).
- 2007 **Master in Bioinformatics**
Sardegna Ricerche, Pula, Italy.
- 2001-2006 **Master in Information Technology**
EPITECH, Paris, France.

WORK EXPERIENCE

- 2018- **Chargé de Recherche (Permanent PI position)**
Inserm / Cancer Research Centre of Lyon, France.
Quantifying, modeling, predicting and controlling cancer evolution and diversity.
- 2016-2018 **Assistant Scientist**
Cancer Research Centre of Lyon / Centre Léon Bérard, Lyon, France.
Analysis of clonal evolution of tumours.
- 2014-2016 **Post-doctoral Research Assistant**
Evolution and Cancer, Dr Trevor Graham's lab, Barts Cancer Institute, Queen Mary University, London, UK.
Analysis of evolutionary processes and dynamics in tumours and pre-malignant conditions.
- 2011-2013 **Post-doctoral Research Assistant**
Translational Cancer Therapeutics, Prof. Charles Swanton's lab, Cancer Research UK, London, UK.
Analysis of intra-tumour heterogeneity and chromosomal instability.

SELECTED PUBLICATIONS in the last 5 years

Muller L, Fauvet F, Chassot C, Angileri F, **Coutant A**, [...], **Martinez P**. EMT-driven plasticity prospectively increases cell-cell variability to promote therapeutic adaptation in breast cancer. *Cancer Cell International*, 2025.

Coutant A, Cockenpot V, Muller L, [...], Ouzounova M*, **Martinez P***. Spatial transcriptomics reveal pitfalls and opportunities for the detection of rare high-plasticity breast cancer subtypes. *Laboratory Investigation*, 2023.

Monteiro L, Da Silva L, Lipinski B, [...], **Martinez P**. Assessing Cell Activities rather than Identities to Interpret Intra-Tumor Phenotypic Diversity and Its Dynamics. *iScience*, 2020.

Tokutomi N, Moyret-Lalle C, Puisieux A, Sugano S, **Martinez P**. Quantifying the fitness of tissue-specific evolutionary trajectories by harnessing cancer's repeatability at the genetic level. *Evolutionary Applications*, 2019.

Beddok A, Velleuer, E, [...], Martinez P, [...], Thariat J. Strategies for early detection and detailed characterization of oral lesions and head and neck squamous cell carcinoma in Fanconi anemia patients. *Cancer Letters*, 2025.

Gatenbee C, Baker AM, Schenck R, [...], Martinez P, [...], Graham TA, Anderson, AR. Immunosuppressive niche engineering at the onset of human colorectal cancer. *Nature Communications*, 2022.

Truchard E, Bertolus C, Martinez P, Thomas E, Saintigny P, Foy JP. Identification of a Gene-Expression-Based Surrogate of Genomic Instability during Oral Carcinogenesis. *Cancers*, 2022.

Bouaoud J, [...], Martinez P, [...], Saintigny P. Unmet Needs and Perspectives in Oral Cancer Prevention. *Cancers*, 2022.

Pommier RM, Sanlaville A, Tonon L, Kielbassa J, Thomas E, Ferrari A, Sertier AS, Hollande F, Martinez P, Tissier A, Morel AP, Ouzounova M & Alain Puisieux. Comprehensive characterization of claudin-low breast tumors reflects the impact of the cell-of-origin on cancer evolution. *Nature Communications*, 2020.

TEACHING

Lectures on Evolutionary trajectories in cancer, Master 2 de Cancérologie, Université Paris Saclay: 2h per year (2020-).

Demonstrations for **structural bioinformatics courses**, undergraduates King's College London: 6 hours (2009).

Lectures on cancer evolution, year 1 medical students Queen Mary University of London: 1 hour per year (2014-2016).

Problem-based learning, Human Science and Public Health module, year 2 medical students, Queen Mary university of London: 14 hours (2016).

Supervision of **8 Master students** in Bioinformatics and **2 undergraduate students** (laboratory technicians).
Main supervisor of **1 PhD student** (started November 2020).

SCIENTIFIC RESPONSIBILITIES

Member of the "**Technological Innovations for Health**" commission for the regional *Cancéropôle Lyon Auvergne Rhône Alpes* Scientific Committee, 2022.

Member of the organising committee for the **Dynamics and statistics of cancer evolution workshop**, Marseille, France, June 2022.

Member of the organising committee for the "**Mathematical Models of Cancer Evolution and Ecology**" symposium during the Mathematical Models of Evolution and Ecology conference, Lyon, France, July 2019.

Member of the organising committee for the **3rd International Symposium of the Cancer Research Centre of Lyon**, Lyon, France, September 2017.

SCIENTIFIC COMMUNICATIONS

Conférence de la Société Française de Médecine Préventive et Personnalisée, 2025, Paris : Tabac, cancer, maladie de Fanconi et évolution somatique de la muqueuse orale. *Invited speaker*

Oral Pre-Malignant Disease Day, 2025, Zagreb, Croatia: Oral mucosa evolutionary dynamics in health and disease. *Invited speaker*

Single Cell Symposium, 2024 Ankara, Turkey: Single-cell analyses of intra-tumour phenotypic heterogeneity and cellular plasticity. *Invited speaker*

RECOMB Satellite Workshop on Computational Cancer Biology, 2023 Istanbul, Turkey: Spatial transcriptomics reveal pitfalls and opportunities for the detection of rare high-plasticity breast cancer subtypes. *Contributed talk*.

Hétérogénéité tumorale : du séquençage de l'ADN aux applications cliniques, 2023, Société Française du Cancer, Paris, France: "Quantification de l'hétérogénéité intra-tumorale". *Invited talk*.

CRCL International symposium, 2019, Lyon, France: "Harnessing cancer's repeatability at the genetic level". *Contributed talk*.

IARI symposium, March 2019, Tokyo, Japan: "Harnessing cancer's repeatability at the genetic level". *Invited speaker*.

The **Cancer Mosaic, Traits, Strategies and Adaptations Workshop**, August 2018, Montpellier, France: "Mapping pre-neoplastic development using contingency-based fitness". *Invited speaker*.

International Society for Evolution, Ecology and Cancer Conference, Tempe AZ, USA, 2017: "Evolution of Barrett's Esophagus through space and time at whole-biopsy and single-crypt levels". *Contributed talk*.

European Conference on Mathematical and Theoretical Biology, July 2016, Nottingham, UK: "Predetermined malignant potential in Barrett's Oesophagus". *Invited speaker (satellite meeting)*.

Nottingham Pathology 2016, June 2016, Nottingham, UK: "Predetermined malignant potential in Barrett's Oesophagus". *Contributed talk*.

Research Progress in Barrett's Oesophagus 9th National Meeting, April 2016, London, UK: "Measuring and modelling evolutionary clonal dynamics in Barrett's Oesophagus to predict cancer risk". *Contributed talk*.

From gene to phenotype: recent advances in biology and medicine workshop, March 2016, Warsaw, Poland. "Pan-cancer analysis of intratumoral heterogeneity". *Invited Speaker*.

Third International Biannual Evolution and Cancer Conference, December 2015, San Francisco, USA: "Predetermined malignant potential in Barrett's Oesophagus". *Contributed talk*.

Cancer Research Centre of Toulouse seminar, November 2015, Toulouse, France: "Quantifying cancer evolution and intratumour heterogeneity". *Invited speaker*.

National Cancer Research Institute (NCRI) Cancer Conference, November 2015, Liverpool, UK: "Longitudinal single cell clonal analysis reveals evolutionary stasis and predetermined malignant potential in non-dysplastic Barrett's Oesophagus". *Contributed talk*.

Evolution and Cancer Conference, Mars 2015, Montpellier, France: "Tracking clonal evolution over time in Barrett's Oesophagus". *Invited speaker*.

Research Progress in Barrett's Oesophagus 8th National Meeting, April 2014, London, UK: "Measuring the pattern of clonal evolution in Barrett's Oesophagus using single-cell FISH data in a large longitudinally-followed cohort". *Contributed talk*.

FUNDING

Projects

2025: Ruban Rose MAESTRO. Team 4 co-leader 75K€ (225K€ total).

2024: Inter-SIRIC collaboration. Team 3 co-leader 50K€ (150K€ total).

2023: Ligue Contre le Cancer « Multi-team project » Project and Team 1 manager €25K (68K total).

2023: Agence Nationale de la Recherche « PRCI ». Project and Team 1 manager €172K (322K total).

2022: Institut National du Cancer « PL-BIO ». Project and Team 1 manager, €224k (412k total).

2021: Institut National du Cancer « PREVBIO ». Team 1 co-manager, €367k (415k total).

2020: Cancéropôle Lyon Auvergne Rhône Alpes « Oncostarter numérique ». Project manager, €40k.

2019: Plan Cancer Single-Cell. Project manager, €360k.

2019: Plan Cancer Maths, Informatics & Cancer. Team 2 manager, €370k (570k total).

Other

2022: Ligue Nationale Contre le Cancer « Subvention colloque » (2,375€).

2019: Cancéropôle Lyon Auvergne Rhône Alpes « Soutien aux manifestations scientifiques » (1,000€).

OTHER

2023 Scientific coordinator of the DEMO clinical protocol (NCT06094829)

2024 Scientific coordinator of the TACO clinical protocol (NCT06425146)